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# Generative AI

**Generative artificial intelligence**, commonly known as **generative AI** or **GenAI**, is a subfield of artificial intelligence that uses generative models to generate text, images, videos, audio, software code (vibe coding) or other forms of data.[1] These models learn the underlying patterns and structures of their training data, and use them to generate new data[2] in response to input, which often takes the form of natural language *prompts*.[3][4]

*Théâtre D'opéra Spatial* (Space Opera Theater, 2022), an image made with Midjourney that won an award at the Colorado State Fair's fine art competition

The prevalence of generative AI tools has increased significantly since the AI boom in the 2020s. This boom was made possible by improvements in deep neural networks, particularly large language models (LLMs), which are based on the transformer architecture. Generative AI applications include chatbots such as ChatGPT, Claude, Copilot, DeepSeek, Google Gemini and Grok; text-to-image models such as DALL-E, Firefly, Stable Diffusion, and Midjourney; and text-to-video models such as Veo, LTX and Sora.[5][6][7]

Companies in a variety of sectors have used generative AI, including those in software development, healthcare,[8] finance,[9] entertainment,[10] customer service,[11] sales and marketing,[12] art, writing,[13] and product design.[14]

Generative AI has been used for cybercrime, and to deceive and manipulate people through fake news and deepfakes.[15][16] Generative AI models have been trained on copyrighted works without the rightholders' permission.[17] Many generative AI systems use large-scale data centers, whose environmental impacts include electronic waste, consumption of fresh water for cooling, and high energy consumption that is estimated to be growing steadily.[18]

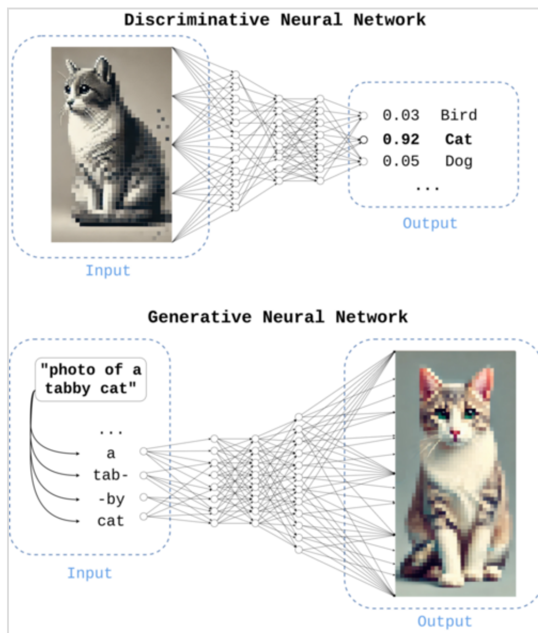
## History

### Early history

The origins of algorithmically generated media can be traced to the development of the Markov chain, which has been used to model natural language since the early 20th century. Russian mathematician Andrey Markov introduced the concept in 1906,[19][20] including an analysis of vowel and consonant patterns in *Eugeny Onegin*. Once trained on a text corpus, a Markov chain can generate probabilistic text.[21][22]

By the early 1970s, artists began using computers to extend generative techniques beyond Markov models. Harold Cohen developed and exhibited works produced by AARON, a pioneering computer program designed to autonomously create paintings.[23] The terms generative AI planning or generative planning were used in the 1980s and 1990s to refer to AI planning systems, especially computer-aided process planning, used to generate sequences of actions to reach a specified goal.[24][25] Generative AI planning systems used symbolic AI methods such as state space search and constraint satisfaction and were a "relatively mature" technology by the early 1990s. They were used to generate crisis action plans for military use,[26] process plans for manufacturing[24] and decision plans such as in prototype autonomous spacecraft.[27]

## Generative neural networks (since the late 2000s)



Above: An image classifier, an example of a neural network trained with a discriminative objective. Below: A text-to-image model, an example of a network trained with a generative objective.

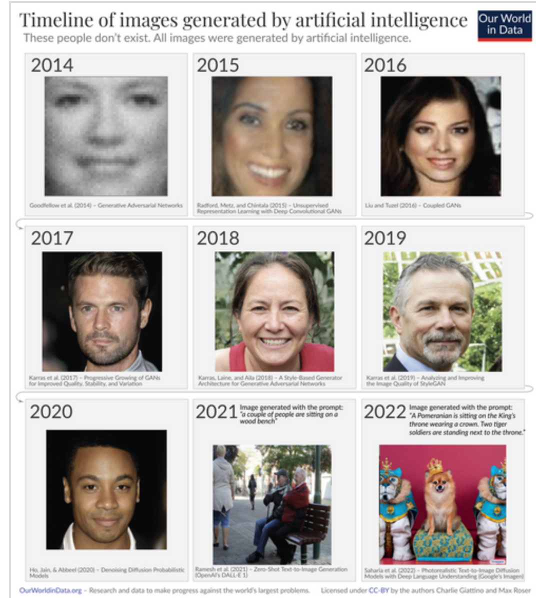
Machine learning uses both discriminative models and generative models to predict data. Beginning in the late 2000s and early 2010s, advances in deep learning led to major improvements in image classification, speech recognition, and natural language processing.[28] Neural networks in this period were typically trained as discriminative models due to the relative difficulty of training generative models.[29]

In 2014, the introduction of models such as the variational autoencoder (VAE) and generative adversarial network (GAN) enabled effective deep generative modeling of complex data such as images.[30][31]

In 2017, the Transformer architecture enabled further advances in generative modeling compared to earlier long short-term memory (LSTM) networks.[32] This led to the development of generative pre-trained transformer (GPT) models, beginning with GPT-1 in 2018.[33]

## Generative AI adoption

In March 2020, the release of 15.ai, a free web application created by an anonymous MIT researcher that could generate convincing character voices using minimal training data, was one of the earliest publicly available uses for generative AI.[34] The platform is credited as the first mainstream service for audio deepfakes.[35][36]



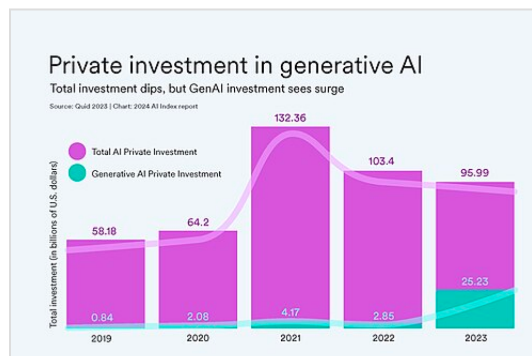

AI generated images have become much more advanced.

In 2021, DALL-E, a closed-source transformer-based generative model developed by OpenAI, drew widespread attention to text-to-image generation.[37]

Other projects, including open-source approaches such as VQGAN+CLIP and DALL·E Mini (later renamed Craiyon), made similar systems more accessible to the public.[38]

Dream by Wombo was released at the end of 2021,[39] followed by the releases of Midjourney and Stable Diffusion in 2022.[40][41]

In November 2022, the public release of ChatGPT popularized generative AI for general-purpose text-based tasks.[42][43][44]




Private investment in AI (pink) and generative AI (green)

In a 2024 survey by marketing research firm Ipsos, Asia–Pacific countries were significantly more optimistic than Western societies about generative AI and show higher adoption rates. Despite expressing concerns about privacy and the pace of change, 68% of Asia-Pacific respondents believed that AI was having a positive impact on the world, compared to 57% globally.[45] According to a survey by SAS and Coleman Parkes Research, as of 2023, 83% of Chinese respondents were using the technology, exceeding both the global average of 54% and the U.S. rate of 65%. A UN report indicated that Chinese entities filed over 38,000 generative AI patents from 2014 to 2023, more than any other country.[46] A 2024 survey by the Just So Soul social media app reported that 18% of respondents born after 2000 used generative AI "almost every day", and that over 60% of respondents like or love AI-generated content (AIGC), while less than 3% dislike or hate it.[47]

By mid 2025 companies were increasingly abandoning generative AI pilot projects as they had difficulties with integration, data quality and unmet returns, leading analysts at Gartner and *The Economist* to characterize the period as entering the Gartner hype cycle's "trough of disillusionment" phase.[48][49]

# Applications

Generative artificial intelligence has been applied across multiple industries for content creation and automation. In healthcare, generative models are used for drug discovery and the generation of synthetic medical data to train diagnostic systems.[50] In finance, they are used for report drafting, data generation, and customer service automation.[51] Media and entertainment industries use generative systems for tasks such as music composition, script development, and image or video generation.[52] Researchers and policymakers have raised concerns regarding accuracy, misuse, and impacts on academic and professional work.[53]

## Text and software code

Large language models (LLMs) are trained on tokenized text from large corpora and are capable of natural language processing, machine translation, and natural language generation.[54]

LLMs can be used as foundation models for a variety of downstream tasks.[55] They can also be trained on source code to generate programs from prompts.[56]

## Audio

In 2016, DeepMind's WaveNet demonstrated that deep neural networks can generate raw audio waveforms.[57] This enabled more realistic speech synthesis compared to earlier approaches. Subsequent systems such as Tacotron 2 demonstrated end-to-end neural text-to-speech generation.[58]

## Images

Generative AI can be used to create visual art.[59] Such systems are trained on image–text pairs. Examples include Stable Diffusion, DALL-E, and Midjourney.[60]

## Video

Generative AI can be used to produce photorealistic videos. Systems such as Runway have demonstrated text-to-video generation capabilities.[61]

## Robotics

Generative models can be used for motion planning and robot control by learning from prior data.[62]

### 3D modeling

Generative models can assist in automating 3D modeling tasks, including generating 3D assets from text or images.[63]

### World models

World models are neural networks designed to learn representations of physical environments, including spatial and dynamic properties.[64] Recent multimodal systems have expanded these capabilities by integrating vision, language, and action into unified models.[65]

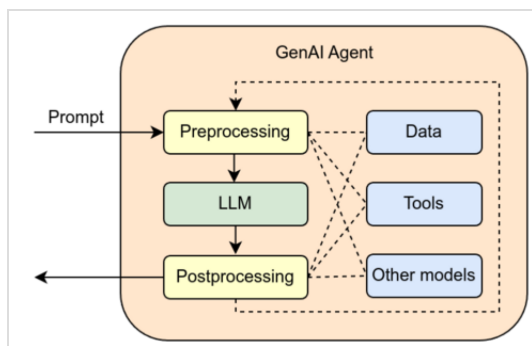
### AI-assisted algorithm generation

Generative AI systems have also been used to assist in the discovery and optimization of algorithms. In this context, AI systems can generate candidate programs, code changes, or mathematical procedures, which are then tested and refined using automated evaluation methods.

In 2023, Google DeepMind introduced AlphaDev, which was used to discover small sorting algorithms that outperformed previously known human benchmarks and have been integrated into the LLVM standard C++ sorting library.[66] In 2025, Google DeepMind introduced AlphaEvolve, an AI system for general-purpose algorithm discovery and optimization. AlphaEvolve uses LLMs to propose code changes, automated evaluators to assess each candidate, and an evolutionary process to iteratively improve algorithms.[67]

## Software and hardware

Generative AI models are used to power chatbot products such as ChatGPT, programming tools such as GitHub Copilot,[68] text-to-image products such as Midjourney, and text-to-video products such as Runway Gen-2.[69] Generative AI features have been integrated into a variety of existing commercially available products such as Microsoft Office (Microsoft Copilot),[70] Google Photos,[71] and the Adobe Suite (Adobe Firefly).[72] Many generative AI models are also available as open-source software, including Stable Diffusion and the LLaMA[73] language model.



Architecture of a generative AI agent

Smaller generative AI models with up to a few billion parameters can run on smartphones, embedded devices, and personal computers. For example, LLaMA-7B (a version with 7 billion parameters) can run on a Raspberry Pi 4[74] and one version of Stable Diffusion can run on an iPhone 11.[75]

Larger models with tens of billions of parameters can run on laptop or desktop computers. To achieve an acceptable speed, models of this size may require accelerators such as the GPU chips produced by NVIDIA and AMD or the Neural Engine included in Apple silicon products. For example, the 65 billion parameter version of LLaMA can be configured to run on a desktop PC.[76]

The advantages of running generative AI locally include protection of privacy and intellectual property, and avoidance of rate limiting and censorship. The subreddit r/LocalLLaMA in particular focuses on using consumer-grade gaming graphics cards[77] through such techniques as compression.

Language models with hundreds of billions of parameters, such as GPT-4 or PaLM, typically run on datacenter computers equipped with arrays of GPUs (such as NVIDIA's H100) or AI accelerator chips (such as Google's TPU). These very large models are typically accessed as cloud services over the Internet.

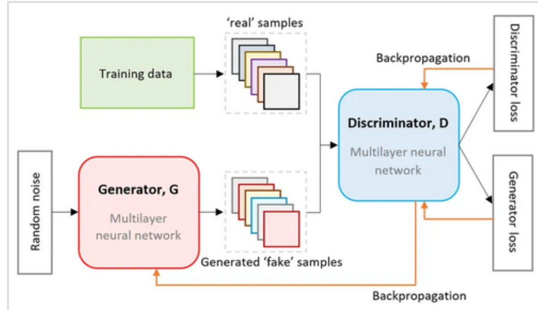
In 2022, the United States New Export Controls on Advanced Computing and Semiconductors to China imposed restrictions on exports to China of GPU and AI accelerator chips used for generative AI.[78] Chips such as the NVIDIA A800[79] and the Biren Technology BR104[80] were developed to meet the requirements of the sanctions.

There is free software on the market capable of recognizing text generated by generative artificial intelligence (such as GPTZero), as well as images, audio or video coming from it.[81] Potential mitigation strategies for detecting generative AI content include digital watermarking, content authentication, information retrieval, and machine learning classifier models.[82] Despite claims of accuracy, both free and paid AI text detectors have frequently produced false positives, mistakenly accusing students of submitting AI-generated work.[83][84]

## Generative models and training techniques

### Generative adversarial networks

Generative adversarial networks (GANs) are a generative modeling technique which consist of two neural networks—the generator and the discriminator—trained simultaneously in a competitive setting. The generator creates synthetic data by transforming random noise into samples that resemble the training dataset. The discriminator is trained to distinguish the authentic data from synthetic data produced by the generator. The two models engage in a minimax game: the generator aims to create increasingly realistic data to "fool" the discriminator, while the discriminator improves its ability to distinguish real from fake data. This continuous training setup enables the generator to produce high-quality and realistic outputs.[85]




Workflow for the training of a generative adversarial network

## Variational autoencoders

Variational autoencoders (VAEs) are deep learning models that probabilistically encode data. They are typically used for tasks such as noise reduction from images, data compression, identifying unusual patterns, and facial recognition. Unlike standard autoencoders, which compress input data into a fixed latent representation, VAEs model the latent space as a probability distribution,[86] allowing for smooth sampling and interpolation between data points. The encoder ("recognition model") maps input data to a latent space, producing means and variances that define a probability distribution. The decoder ("generative model") samples from this latent distribution and attempts to reconstruct the original input. VAEs optimize a loss function that includes both the reconstruction error and a Kullback–Leibler divergence term, which ensures the latent space follows a known prior distribution. VAEs are particularly suitable for tasks that require structured but smooth latent spaces, although they may create blurrier images than GANs. They are used for applications like image generation, data interpolation and anomaly detection.

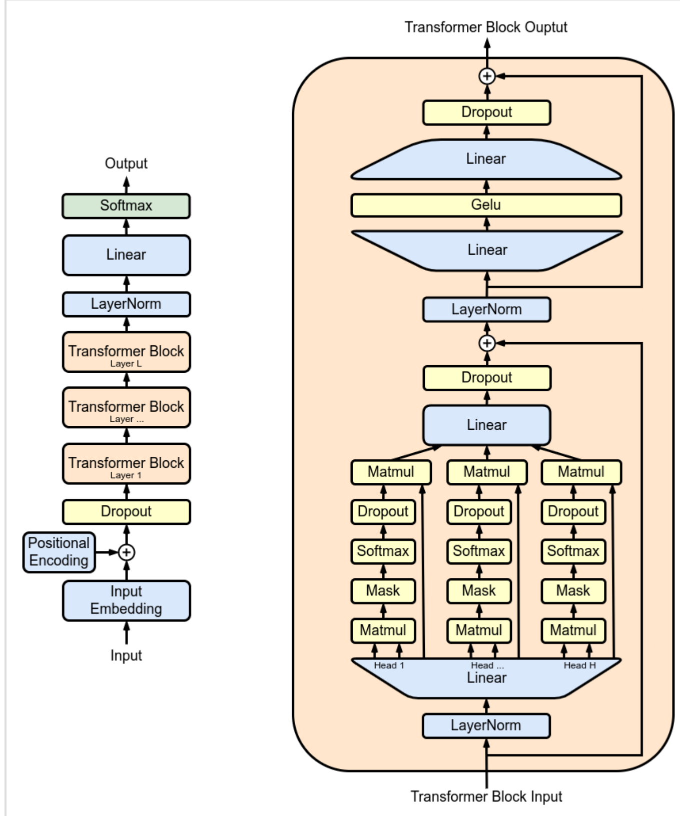
Comparison between images generated by a VAE (left) and a GAN (right). VAEs tend to produce smoother but blurrier images due to their probabilistic decoding.

## Transformers

Transformers became the foundation for the generative pre-trained transformer (GPT) series developed by OpenAI, replacing traditional recurrent and convolutional models.[87]

The self-attention mechanism enables the model to determine the relative importance of each token in a sequence when predicting the next token, thereby improving contextual understanding. Unlike recurrent neural networks, transformers process tokens in parallel, which improves training efficiency and scalability.[88]

Transformers are typically pre-trained on large corpora using self-supervised learning and then fine-tuned for specific tasks.[89]




The full architecture of a GPT model

# Law and regulation

In the United States, a group of companies including OpenAI, Alphabet, and Meta signed a voluntary agreement with the Biden administration in July 2023 to watermark AI-generated content.[90] In October 2023, Executive Order 14110 applied the Defense Production Act to require all US companies to report information to the federal government when training certain high-impact AI models.[91][92]

In the European Union (EU), the proposed Artificial Intelligence Act includes requirements to disclose copyrighted material used to train generative AI systems, and to label any AI-generated output as such.[93][94]

In China, the Interim Measures for the Management of Generative AI Services introduced by the Cyberspace Administration of China regulates any public-facing generative AI. It includes requirements to watermark generated images or videos, regulations on training data and label quality, restrictions on personal data collection, and a guideline that generative AI services must "adhere to socialist core values".[95][96]

## Copyright

### Training with copyrighted content

Generative AI systems such as ChatGPT and Midjourney are trained on large, publicly available datasets that include copyrighted works. AI developers have argued that such training is protected under fair use, while copyright holders have argued that it infringes their rights.[97]

Proponents of fair use training have argued that it is a transformative use and does not involve making copies of copyrighted works available to the public.[97] Critics have argued that image generators such as Midjourney can create nearly-identical copies of some copyrighted images,[98] and that generative AI programs compete with the content they are trained on.[99]

As of 2024, several lawsuits related to the use of copyrighted material in training are ongoing. Getty Images has sued Stability AI over the use of its images to train Stable Diffusion.[100] Both the Authors Guild and The New York Times have sued Microsoft and OpenAI over the use of their works to train ChatGPT.[101][102]

### Copyright of AI-generated content

A separate question is whether AI-generated works can qualify for copyright protection. The United States Copyright Office has ruled that works created by artificial intelligence without any human input cannot be copyrighted, because they lack human authorship.[103] Some legal professionals have suggested that *Naruto v. Slater* (2018), in which the U.S. 9th Circuit Court of Appeals held that non-humans cannot be copyright holders of artistic works, could be a potential precedent in copyright litigation over works created by generative AI.[104] However, the office has also begun taking public input to determine if these rules need to be refined for generative AI.[105]

In January 2025, the United States Copyright Office (USCO) released extensive guidance regarding the use of AI tools in the creative process, and established that "...generative AI systems also offer tools that similarly allow users to exert control. [These] can enable the user to control the selection and placement of individual creative elements. Whether such modifications rise to the minimum standard of originality required under Feist will depend on a case-by-case determination. In those cases where they do, the output should be copyrightable"[106] Subsequently, the USCO registered the first visual artwork to be composed of entirely AI-generated materials, titled "A Single Piece of American Cheese".[107]

## Concerns

The development of generative AI has raised concerns from governments, businesses, and individuals, resulting in protests, legal actions, calls to pause AI experiments, and actions by multiple governments. In a July 2023 briefing of the United Nations Security Council, Secretary-General António Guterres stated "Generative AI has enormous potential for good and evil at scale", that AI may "turbocharge global development" and contribute between $10 and $15 trillion to the global economy by 2030, but that its malicious use "could cause horrific levels of death and destruction, widespread trauma, and deep psychological damage on an unimaginable scale".[108] In addition, generative AI has a significant carbon footprint.[109]

## Societal impacts

### Academic honesty

Generative AI can be used to generate and modify academic prose, paraphrase sources, and translate languages. The use of generative AI in a classroom setting has challenged traditional definitions of academic plagiarism, leading to a "cat-and-mouse" dynamic between students using AI and institutions attempting to detect it.[110] In the immediate wake of ChatGPT's release, many school districts and universities issued temporary bans on the technology, though many institutions have since moved toward policies of managed integration.[110] However, the implementation of these policies often lacks clarity. Research suggests that the burden of interpreting "acceptable use" frequently falls on individual students and teachers, creating an environment where academic honesty becomes difficult to define and enforce.[111]

A commonly proposed use for teachers is grading and giving feedback. Companies like Pearson and ETS use AI to score grammar, mechanics, usage, and style, but not for main ideas or overall structure.[112] The National Council of Teachers of English stated that machine scoring makes students feel their writing is not worth reading.[113] AI scoring has also given unfair results for students from different ethnic backgrounds.[114]

### Fears over job losses

From the early days of the development of AI, there have been arguments put forward by ELIZA creator Joseph Weizenbaum and others about whether tasks that can be done by computers actually should be done by them, given the difference between computers and humans, and between quantitative calculations and qualitative, value-based judgements.[116] In April 2023, it was reported that image generation AI has resulted in 70% of the jobs for video game illustrators in China being lost.[117][118] In July 2023, developments in generative AI contributed to the 2023 Hollywood labor disputes. Fran Drescher, president of the Screen Actors Guild, declared that "artificial intelligence poses an existential threat to creative professions" during the 2023 SAG-AFTRA strike.[119] Voice generation AI has been seen as a potential challenge to the voice acting sector.[120][121]

However, a 2025 study concluded that the US labor market had so far not experienced a discernible disruption from generative AI.[122] Another study reported that Danish workers who used chatbots saved 2.8% of their time on average, and found no significant change in earnings or hours worked.[123]